Lishu Zhang

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Employment

2025–present Postdoctoral Researcher, CREST, ENSAE Paris, Institut Polytechnique de Paris

Education

2021–present Ph.D. in Economics, Tilburg University
Defense expected: Spring 2027

2019–2021 Research Master in Economics, Tilburg University, with distinction

2017–2019 Master of Science in Economics, Stockholm School of Economics

2013–2017 Bachelor of Economics, Renmin University of China

Research Interests

Financial Economics, Environmental Economics, International Economics
Focus: Climate Finance, Cross-border Capital Flows, Production Networks, Critical Raw Materials

Working Papers

Measuring Critical Raw Material Dependence (w/ Peter Tankov and Olivier David Zerbib) ***(Job Market Paper)***

The supply of critical raw materials is increasingly exposed to geopolitical disruption, yet little is known about which firms ultimately bear these shocks or how they adjust. We develop a firm-level measure of structural material dependence by tracing material requirements through production paths and firms' product portfolios. Technical evidence identifies the relevant product links, and input cost shares translate them into marginal-cost exposure. The resulting measure covers 12,478 firms across 55 materials. Dependence varies substantially within narrow industries, while filings and specialist reports cover less than half of measured dependence in their matched samples. The cost response provides outcome-based validation of the measure: a unit of predicted pressure is associated with a 1.02 percent increase in cost of goods sold relative to initial sales, closely matching the implied scale. Higher material-price pressure is followed by lower operating cash flow and greater external financing, without an immediate decline in capital expenditure. These responses are stronger for price shocks identified from supply-tightening events. More dependent firms also exhibit higher return volatility and downside risk, especially when they rely on more volatile materials. A 10 percent event-driven price increase across the measured materials implies $17.8 billion of operating cash flow at risk, 48 percent in relationships absent from specialist reports.

Resistance and Arbitrage: International Trade in Brown Loans

Yihong Xia Best Paper Award, CICF 2025.

I develop a novel measure of carbon sensitivity in lending to assess reductions in portfolio exposure to brown assets. Using syndicated loan data, I show that countries with greater resistance to brown lending, proxied by economic development, experience faster shifts in the sectoral composition of loan portfolios. The decarbonization

is driven primarily by domestic credit reallocation. I find consistent evidence of risk transfers to less regulated lenders and foreign countries, indicating arbitrage and incomplete regulations. Furthermore, lenders' climate risk-taking and transfer behaviors vary sharply by syndicate role, loan type, and specialization. The existence of international trade in brown loans has important implications for supervisory evaluation. Using the European Central Bank's climate guide, I show that accounting for regulatory leakage reveals effects contrary to common wisdom.

Who holds brown equity? Economic development, divestment, and environmental outcomes (w/ Harald Benink, Harry Huizinga, and Louis Raes) (Under review)

Using global equity-ownership data for 2002–2021 covering 3,559 firms and 111 investor countries, we show that investors in more developed countries hold less carbon-intensive portfolios. This income gradient appears within varying investor types and remains after accounting for return chasing and financial development, consistent with environmental concerns being more strongly reflected in richer-country portfolios. Greater ownership by poorer-country investors predicts higher subsequent emissions and ESG incidents. Exploiting waves of fossil-fuel divestment commitments, we construct a shift-share measure using firms' predetermined ownership. Rich-country divestment pressure shifts ownership toward poorer countries and worsens environmental performance, reduced-form effects consistent with an ownership channel.

Does ESG information shape control? Evidence from takeovers

When two firms combine into one, the allocation of control determines who manages their assets. We study whether ESG information shapes control using 7,914 completed deals announced from 2003 to 2025. We estimate that ESG rating availability increases the rated firm's predicted probability of taking control by 35–37 percentage points between two otherwise equal firms. Where both firms are rated, control tends to go to the higher-scoring firm, most strongly on the Environmental pillar across borders. The information advantage is larger under strong audit standards and smaller where corporate reporting is widespread. Deal-level announcement returns do not differ significantly across combinations of acquirer and target rating status, but after 2016 a rated target bought by an unrated firm earns less than an unrated target bought by a rated firm.

Work in Progress

Localized Impacts of Transition Risks (w/ Thibaud Barreau, Jakob Kellermann, Peter Tankov, and Olivier David Zerbib)

We develop a facility-level measure of transition risk based on the profit losses that would arise if carbon prices moved toward local social-cost-of-carbon benchmarks. The framework combines plant emissions, effective policy coverage, abatement options, and uncertain repricing dates. By matching facilities to detailed carbon-pricing rules, it captures variation that national headline prices and firm-level emissions measures miss. The next stage will connect this exposure map to the local economic and environmental effects of realized transition shocks.

Teaching Experience

Spring 2022–2025	Teaching Assistant, *Macroeconomics 3*, PhD level, Tilburg University
Fall 2020–2025	Teaching Assistant, *Economics of the European Union*, undergraduate level, Tilburg University
Fall 2020–2022	Teaching Assistant, *Economics of Banking and Finance*, undergraduate level, Tilburg University
Fall 2018	Teaching Assistant, *International Economics*, undergraduate level, Stockholm School of Economics

Grants, Scholarships, and Awards

2026	AFA Travel Grant, American Finance Association
2025	Yihong Xia Best Paper Award, CICF
2019–2021	Koopmans Scholarship, Tilburg University
2015	National Scholarship, Renmin University of China
2014	Outstanding School Cadre, Renmin University of China

Seminar & Conference Presentations

2026	AFA Poster Session; Junior Finance Research Day, CREST; CEMRACS; HEC Paris Finance PhD Workshop
2025	CICF; AsianFA PhD Consortium; PhD-Economics Virtual Seminar
2023–2024	FMA; Canadian Economics Association; DEARE, Utrecht University; ESSFM, Gerzensee; Imperial College PhD Conference; KU Leuven; Tilburg University (Finance, Economics)

University Service

2021–2023	Member, Tilburg PhD Platform (TiPP)
2019–2020	Representative of Research Master's in Economics, Graduate Students' Society (GSS)

Computer Skills

STATA, R, MATLAB, Python, LaTeX.

Languages

English, French (intermediate), Mandarin

References

Prof. Harry Huizinga
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